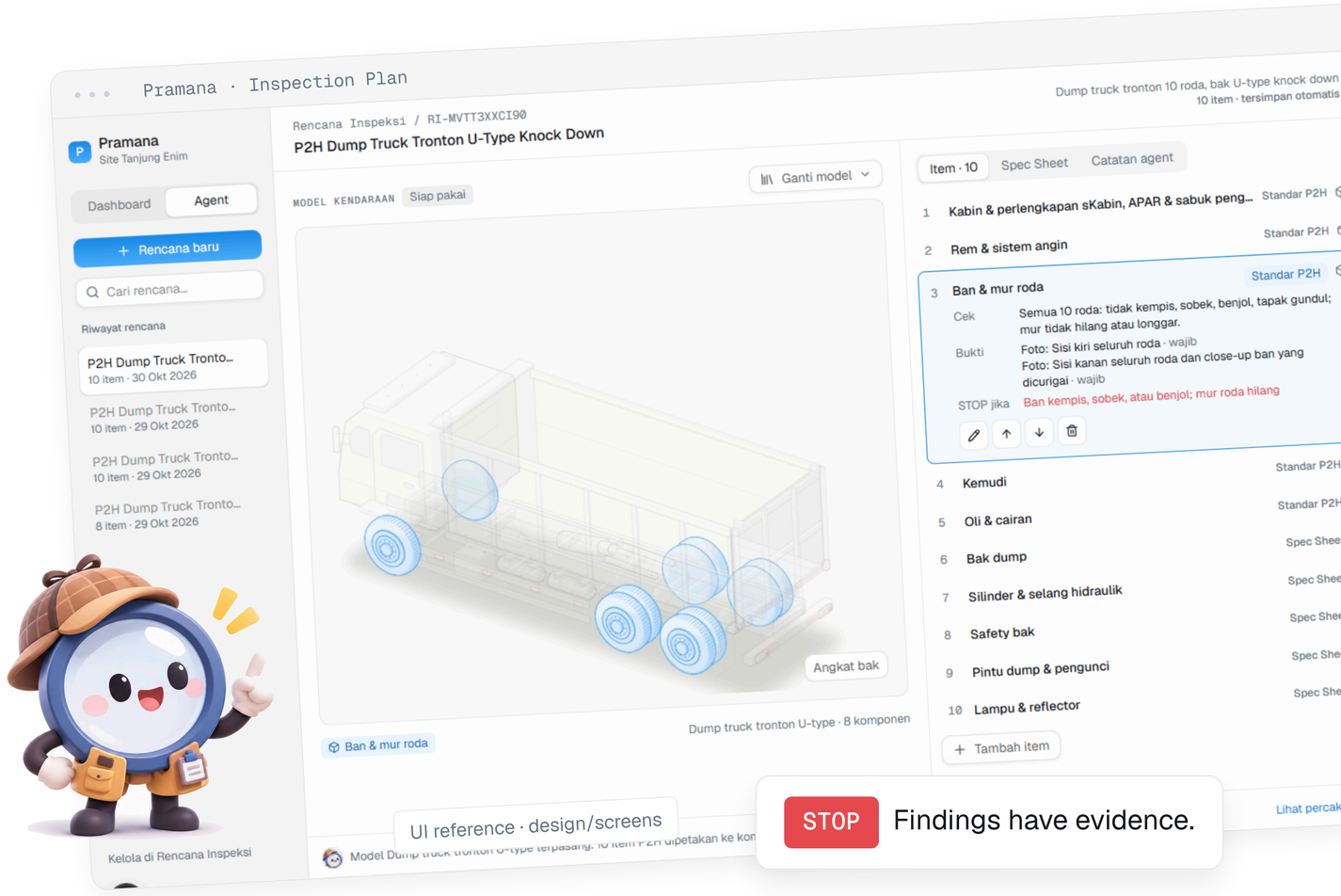
Your P2H companion before each shift

P2H backed by evidence

Pramana helps inspectors check each vehicle, collect evidence, and follow up on findings.

Spec Sheet → Photo evidence → Action



P2H: Daily Inspection and Maintenance · Initial focus: dump trucks

p2h.apta.works ↗

Problem & Solution Fit

A checkmark for “good” does not show what was actually inspected

✓

All good.

Problem scenario from the PRD.
A completed form does not establish evidence quality.

01 **Vehicle-specific components can be missed.**

A generic plan may miss wheels, dump body safety, or the vehicle’s hydraulics.

02 **A photo may lack the detail needed.**

The fire extinguisher is visible from a distance, but its label and details are unreadable.

03 **Findings need follow-up.**

Supervisors need to see vehicles marked STOP and repairs still awaiting action.

Problem basis: PRD §1. Product needs hypothesis, without user research or prevalence data yet.

The solution workflow

Vehicle specifications guide action with context

Spec Sheet + P2H Standard

Create an Inspection Plan.

The supervisor reviews components, required evidence, and STOP conditions.

Inspector + Lumo

Inspect the vehicle. Complete the evidence.

The conversation requests photos and clarification for each component.

Inspection Report

Store findings with their photos.

The supervisor can review the status and evidence.

Supervisor + mechanic

Follow up through a Work Order.

Each repair has an assigned mechanic, a deadline, and a handling status.

Lumo helps interpret input.

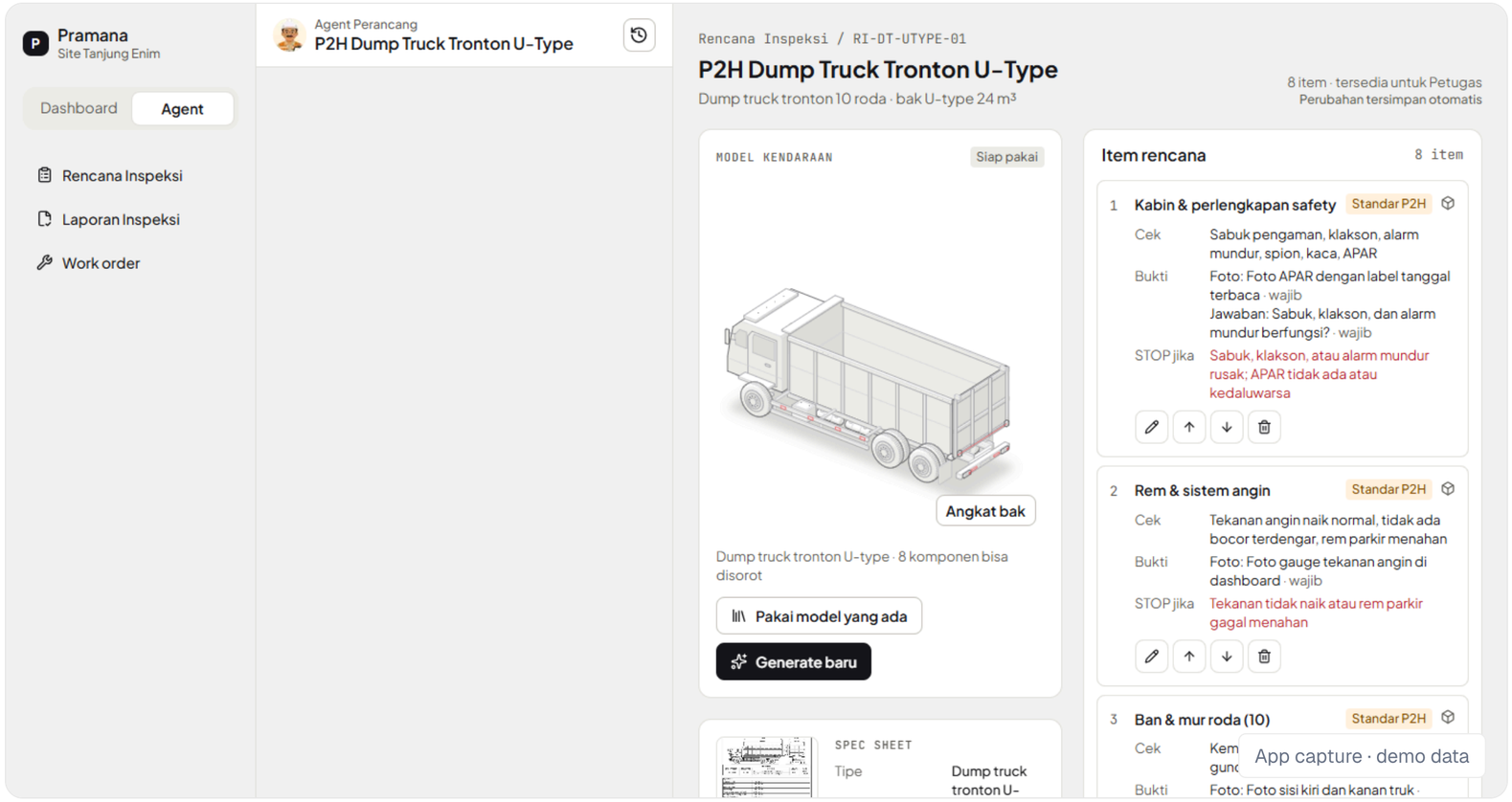
Code enforces sequence, completeness, and report status.

People control actions

Sources: PRD §4–6 · CONTEXT.md · contracts and rules in the implementation repository.

Spec Sheet → Inspection Plan

Each inspection fits the vehicle



Specifications meet the P2H Standard.

Vehicle components combine with checks required by the standard.

Each item defines evidence and STOP.

Inspectors get specific instructions for each check.

Tires & wheel nuts

Check the wheels. Request photos. Identify STOP conditions.

Open Inspection Plans ↗

PRD §4.1 · Example plan for a U-type dump truck. Each item's source remains available for review.

A conversation that requests relevant evidence

Unclear evidence?
A clearer photo while still on site



1 · Distant photo | 2 · Retake photo | 3 · Tire bulge

A distant photo prompts a request for clearer evidence.

Lumo
Inspection companion

Photo retake needed

I can’t read the label yet. Could you take a closer photo?

Recording principle

A visible component ≠ enough information.

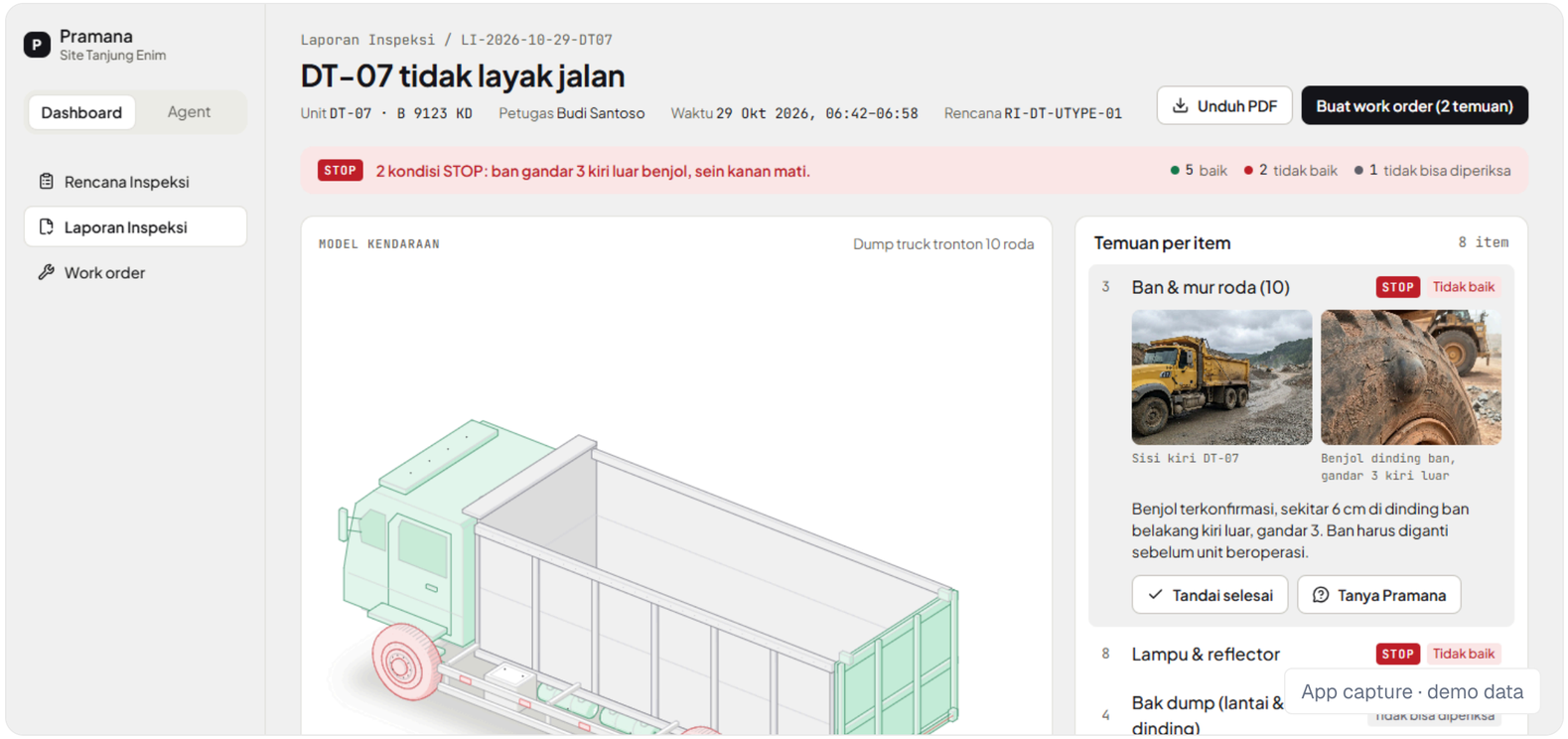
Interactive simulation using PRD scenarios and sample photos. Responses are scripted, without a live vision call.

Open Inspector view ↗

Report → Work Order → Resolution

Findings continue beyond the report

Inspection Report | Work Order · UI reference | Ask Lumo · UI reference



LI-2026-10-29-DT07 · DT-07

Findings keep their evidence.

One Inspection Report shows photos, components, notes, and STOP conditions.

Open demo app ↗

Ask Lumo uses report/WO sources and fallback answers as specified in the P3 design.

The UI shows sample data. Closing a WO records handling. Vehicle roadworthiness requires another inspection.

Build Quality

Agents assess input.
Code enforces the rules

Situation	Enforced behavior	Evidence in the Lumo implementation
A required photo is missing.	Reject the item result and request evidence first.	`evidenceProblem()` `lib/rules.ts`
Some items are incomplete.	Block report submission.	`nextItem()` `submit_report`
The Analyst agent fails.	Provide a labelled fallback answer from the data.	`jawabCadangan()` `lib/agents/analis.ts`
A WO closes or reopens.	Update the finding’s handling status too.	`updateWorkOrder()` `lib/work-order.ts`

The supervisor controls actions. Ask Lumo only reads data.

Reviewed source: lumo-design branch, c70e025. Source review and end-to-end deployment tests are distinct evidence.

Insight & Originality

Components, evidence, and actions share the same context

Vehicle context

The plan follows the vehicle type.

The Spec Sheet complements the P2H Standard. Items link to the Vehicle Model.

Inspection context

Questions follow evidence quality.

Unclear photos prompt more specific requests while the inspector is still on site.

Follow-up context

Findings stay linked to their sources.

Work Orders and recommendations retain the report ID and the affected component.



The design preserves context throughout the workflow, linking inspection requirements to the repairs needed.

Real-World Value

Clearer checks in the field.
Findings ready for follow-up

Inspector

Knows which components to check and which photos to collect.

Supervisor

Reviews STOP findings with evidence and vehicle context.

Mechanic

Receives a Work Order linked to the finding.

<30 seconds

Target: create a plan with ≥8 items.

<4 minutes

Target: 8-item P2H on a phone.

Measure during the pilot:
evidence completeness, review time, and finding follow-up.

These are PRD targets. They are not measured results or claims of operational impact.

Target source: PRD §9. Validation plan: compare Pramana with the team's existing P2H process.

Alignment with Build Direction

AI that understands the work of P2H

Primary direction

Deep Domain AI

A concrete domain: dump truck components, the P2H Standard, required evidence, STOP conditions, and finding follow-up.

Plans by vehicle type Evidence by component STOP

Supporting direction

AI-Native Products & Operations

Adaptive conversations guide evidence collection and produce structured reports.

Autonomy stays within inspection assistance. People still approve plans and decide on corrective actions.

Project alignment reflects the team's assessment. Build direction names follow the official hackathon guide.

Use of Codex

Codex gets context, contracts, and completion criteria

Codex · work instructions from the repo

```
Work through P0 tickets in order.
Read PRD, CONTEXT, and DESIGN_SYSTEM.
Keep evidence rules in code.
Run tsc, build, and UI verification.
Make small commits after each ticket.
```

P0 tickets ↗ | Definition of Done ↗ | Design system ↗

Domain context before coding.

Repo documents define terminology, data sources, and task boundaries.

Tickets define verifiable work.

P0 → P3. Each scope includes data contracts and verification criteria.

Key decisions remain reviewable.

Deterministic rules, ADRs, reference screenshots, and source changes are available.

Available evidence: workflow and repo artifacts.
Codex session logs are not attached yet.

Instructions summarized and translated from tickets/README.md. Not an execution log or a claim about Codex's share of the code.

Evidence-based P2H

Beyond checkmarks, evidence that supports action

Start with one vehicle.
Keep its context through follow-up.

Open app demo ↗ | 6-minute script ↗



p2h.apta.works ↗

Supervisor on laptop · Inspector on phone

Plans fit the vehicle Evidence per component Findings link to Work Orders

Q&A · Source branch lumo-design

Flexible conversations. Structured records

Interface

Supervisor
Inspector
Vehicle Model

↓

`Plans / conversations`
`Photos / reports / WOs`

Agents by responsibility

Planner · Inspector
Model Builder · WO Drafter
Read-only Analyst

↓

`set_items · record_item`
`submit_report · sumber[]`

Rules & storage

Evidence validation
Progress and verdicts
Handling status

↓

`Data contracts`
`SQLite · local traces`

Ready-made Vehicle Models provide a fallback when generation is unavailable. The Analyst can fall back to data-based answers.

Basis: ADRs 0004–0007 and lumo-design source. Code was reviewed. Not all failure paths were tested on the deployment.

Q&A · Reviewable verification

Evidence, targets, and scope

The deployment is accessible.

Accessible pages and captures appear in capture.json ↗. These checks only navigated and read existing data.

The Lumo design covers more features.

Images in design/screens are UI references. Deployment labels and feature coverage may differ from those references.

Targets still need measurement.

A ≥8-item plan in <30 seconds and 8-item P2H in <4 minutes come from the PRD. No accuracy, savings, or incident reduction claims.

The demo has a defined scope.

The focus is dump truck P2H. Login, multiple sites, offline mode, voice, and push notifications sit outside the PRD scope.

Acceptance tests still to demonstrate.

Missing photos block results. Empty items block submission. STOP findings stay visible. WOs synchronize. Fallback answers cite sources.

Next validation step: a pilot.

Compare Pramana with the current process: evidence completeness, repeat clarification, review time, and finding follow-up.

Sources: PRD §3, §9, §11 · app captures · implementation source. Presentation checks are separate from application tests.

Q&A · Arguments and evidence

Six criteria. One product story

Criterion	Argument / evidence shown	Location
Problem & Solution Fit	P2H evidence gaps lead to vehicle-specific plans and photo conversations.	Problem · Workflow · Plan
Build Quality	App UI, evidence validation in code, progress, WO synchronization, and fallback answers.	Demo · Reliability
Insight & Originality	Component context connects evidence requirements to follow-up.	Evidence · Distinction
Real-World Value	Concrete benefits for inspectors, supervisors, and mechanics, with a pilot measurement plan.	Follow-up · Benefits
Alignment with Build Direction	Deep Domain AI as the primary direction. AI-Native Products & Operations as support.	Build direction
Use of Codex	PRD, P0–P3 tickets, contracts, ADRs, and Definition of Done. Session attribution is not attached yet.	Codex · Repo artifacts

Rubric follows the supplied judging-criteria.jpg. No scoring weights are assumed.